Wenhui Li Ph.D.

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Summary

- 10+ years of expertise in multiscale modeling and simulation across atomistic, mesoscopic, and continuum scales
- Skilled in physics-informed machine learning models across a range of scientific and engineering domains
- Proven record of scientific innovation with 27 publications (800+ citations) and interdisciplinary collaborations

Skills

- **Domain Knowledge**: Computational Soft Materials, Solid/Fluid Mechanics, Interface Science
- **Programming**: Python (PyTorch, TensorFlow, pandas, scikit-learn, SciPy), C, C++, Bash
- **Machine learning**: LLM, Gen AI (GAN & VAE), GNN, Unsupervised learning
- **Simulation & Modeling**: Atomistic (Molecular Dynamics (MD) & Monte Carlo (MC)), Mesoscopic (Coarse-Grained Modeling), Continuum (Finite Difference/Volume/Element Methods)
- **Tools & Platforms**: HPC (SLURM), Linux, Git, AWS
- **Soft Skills**: Fast Learner, Detail-oriented, Multitasker, Collaborator, Professional Communicator

Professional Experience

Postdoctoral Researcher *Sep. 2022 – Present*
Indiana University, Bloomington | *Toyota*, Princeton, IN, USA

- Developed Graph Neural Networks to accelerate MD simulations, improving computational speed by ~30%
- Built and deployed ML-based surrogate models for Toyota's paint defect prediction, achieving a ~50% reduction in manual inspection
- Contributed to the AI-based energy analytics project (AnalytiXIN) for Indiana State manufacturers, reducing energy consumption by 20–30%
- Partnered with experimentalists to elucidate nanoparticle encapsulation mechanisms via coarse-grained MD and PMF analysis, contributing to the rational design of cargo loading encapsulation
- Uncovered molecular shear thinning mechanisms in lubricants via MD simulations and ML-assisted analysis

Postdoctoral Researcher *Jul. 2021 – Jul. 2022*
University of Alberta, Edmonton | *Suncor Energy & Syncrude*, Calgary, AB, Canada

- Simulated water/oil/asphaltene interfaces under electric fields, revealing field-induced disruption of asphaltene $\pi - \pi$ stacking and adsorption, with salt ions enhancing interfacial stability
- Investigated brine droplet behavior in oil/asphaltene under electric fields via MD simulations, showing that ion type, concentration, and asphaltene adsorption jointly govern droplet stability and migration—offering insights into electrostatic demulsification mechanisms

PhD Student *Aug. 2017 – Jul. 2021*
University of Alberta, Edmonton, AB, Canada

- Investigated how surfactants, co-surfactants, salt ions, and nanoparticles affect the structural and dynamic properties of hydrocarbon–water interfaces, highlighting the synergistic role of co-surfactants in reducing interfacial tension and clarifying the debated effects of nanoparticles and ion concentrations
- Investigated CO_2 solubility in confined environments, revealing how surface chemistry, pore size, and salt ions jointly govern "over-solubility" or "under-solubility," demonstrating the potential of CO_2 geo-sequestration
- Developed in-house codes to characterize amorphous nanoporous media and simulate gas transport, uncovering how porosity, pore connectivity, size, and tortuosity influence diffusion—providing insights for optimizing shale gas recovery.

Master Student *Sep. 2014 – Jul. 2017*
China University of Geosciences at Beijing, Beijing, China

- Developed and validated a mathematical model for foamy heavy oil transport in porous media, demonstrating superior performance compared to existing models

- Constructed and validated a mathematical model capturing the temporal evolution of pressure fields with source/sink terms

Education

Ph.D. in Civil and Environmental Engineering — *Jul. 2021*
University of Alberta, Edmonton, AB, Canada

M.S. in School of Energy — *Jul. 2017*
China University of Geosciences, Beijing, China
Outstanding Graduate Thesis (top 5%)

B.S. in School of Energy — *Jul. 2014*
China University of Geosciences, Beijing, China
Graduated with Distinction (Ranked 3/67)

Selected Publications & Presentations

Authored 27 peer-reviewed publications, including 12 as first author, with 800+ citations
Delivered 15 presentations at leading conferences, including the APS, AIChE, and Society of Rheology
(Partial list shown; full list available on my Google Scholar)

- Industrial Energy Forecasting Using Dynamic Attention Neural Networks. *Energy and AI*, 20 (2025): 100504.
- Data-Driven Discovery of Molecular Structure Patterns in Shear Thinning of Lubricants. *95th Society of Rheology*, Austin, TX, USA, Oct. 14-17, 2024.
- Electrostatics-Driven Peptide-Directed Encapsulation of Nanoparticles into Protein Cages. *APS March Meeting*, Minneapolis, MN, USA, Mar. 3-8, 2024.
- Dependence of Methane Transport on Pore Informatics in the Amorphous Nanoporous Kerogen Matrix. *Langmuir*, 40 (2023): 687-695.
- Molecular Dynamics Simulation on Water/Oil/Asphaltene Interface Subjected to Electric Field. *Journal of Colloid and Interface Science*, 628 (2022): 924-934.

Leadership & Service

- Led the project “Interfacial Behaviors During the Development of Unconventional Gas Reservoirs”; authored the research proposal and secured USD 4,000 in funding from May 2019 to May 2020.
- Mentoring 5 graduate/undergraduate students, resulting in joint publications and successful project delivery
- Co-chaired the “Data-driven Rheology” session at *Society of Rheology Meeting*, 2024
- Reviewed 20+ papers for Physical Review Letters, Langmuir, Fuel, Chemical Engineering Journal, etc.
- Guest Editor for *Geofluids* special issue on Multiscale and Multiphysics Approaches to Fluids Flow in Unconventional Reservoirs, 2022

Honors & Awards

Future Energy Systems Travel Fund, University of Alberta	*2019*
Graduate Student Association Academic Travel Grant, University of Alberta	*2018*
Graduate Recruitment Scholarship (CAD 10,000), University of Alberta	*2017*
National Scholarship (2×, top 3%, CNY 20,000), Ministry of Education, China	*2015 & 2016*